COURSE SLIDES

Generative AI for SEO

Course Code: C11

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v1.0 · 28 July 2026

COURSE ADMINISTRATION

Welcome & Housekeeping

TRAINERS LISTED FOR C11

Course Trainer Profiles

1

Ray Teoh Tham Kim

Digital marketing leader with regional business-growth and training experience.

2

Patrick Foo

SEO practitioner and adult educator specialising in practical website optimisation.

3

Janice Ong

Business, e-commerce and digital marketing leader with extensive training experience.

ICE-BREAKER

Let's Know Each Other

1. Your name, organisation and role.
2. A page, product or service you would like more people to find.
3. One SEO task you currently do manually.
4. Which assistant you can access today: ChatGPT, Claude or Gemini.

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. Learn by doing — try every hands-on lab.

COURSE MATERIALS

Access Course Materials

1. Open the C11 course page or the trainer-supplied Drive folder
2. Open the courseware links for this course
3. Download the Slides, Learner Guide and Lesson Plan
4. Open the Activities folder for all six labs
5. Keep the files together as you work

Course page: tertiarycourses.com.sg/generative-ai-for-seo.html

SCHEDULE

Lesson Plan — 1 Day, 7.5 instructional hours

Course sequence

- Day 1 — Research, create, optimise and verify one evidence-led SEO workflow

– Topic 1: Getting Started with Generative AI for SEO (Labs 1–3)

– Topic 2: Creating and Optimising SEO Content with Generative AI (Labs 4–6)

Day format

- Daily timing

– 9:30 am – 6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)

- Instructional time

– 7.5 hours · 6 connected labs

- Learning arc

– Concept → demonstrate → build → verify → discuss → recap

WHAT YOU'LL ACHIEVE

Learning Outcomes

1

Set Up & Prompt

Explain how SEO and generative AI work together, set safe working boundaries and use a structured prompt-and-review method.

2

Research & Plan

Build an evidence-led keyword universe, topic clusters and a search-intent content plan with AI assistance.

3

Create & Optimise

Create an SEO content brief, outline, title, meta description and on-page draft that remain grounded in approved sources.

4

Audit & Verify

Audit and improve an existing page for content, on-page and technical SEO, then verify quality against Google guidance.

LEARNING APPROACH

How You'll Learn

1. Trainer demonstrates the concept
2. You build it yourself in the hands-on lab
3. Verify with the lab's 'Test it' step
4. Discuss and troubleshoot together
5. Recap against the learning outcomes

CORE CONCEPTS

The AI-Assisted SEO Operating System

OVERVIEW

Keep Four Kinds of Information Separate

1. **Observed evidence**
   Source-linked business facts, query data, page code and performance exports.

2. **AI suggestions**
   Proposals to compare or refine; never treated as measurements or guarantees.

3. **Human decisions**
   Approved audience, priority, page purpose, wording, release and stop decisions.

4. **Unknowns**
   Missing or uncertain information that stays visible until it can be resolved.

THE BUILDING BLOCKS

G-C-C-S-O-R Prompt Framework

1 **Goal**
One task and the decision the output must support.

2 **Context**
Audience, offer, page job and business situation.

3 **Constraints**
Scope, exclusions, tone, privacy and policy boundaries.

4 **Sources**
Delimited evidence the assistant may use.

5 **Output**
A precise schema, length or example.

6 **Review**
Claim checks, missing evidence and human acceptance criteria.

C11 OPERATING PRINCIPLE

AI accelerates the workflow; evidence and judgment determine the quality.

Never ask a model to invent demand, guarantee rankings or replace the human editor.

COURSE ROADMAP

What You Will Build

Research system

- AI-ready brief
- keyword evidence
- topic clusters

Content system

- intent plan
- brief and outline
- metadata and page draft

Quality system

- page audit
- claim ledger
- prioritised improvement plan

The Learning Arc in Every Lab

1. Open the checkpoint and approved source files.
2. Use a bounded prompt to transform—not invent—evidence.
3. Make a human decision and save it in the working artifact.
4. Run the Test It check and keep the checkpoint for the next lab.

BEFORE ANY PAGE CHANGE

The Human Review Gate

1 **Evidence**
Can every material claim and metric be traced?

2 **Intent**
Does the page help the visitor complete the intended job?

3 **Quality**
Is the contribution accurate, useful, original and readable?

4 **Technical**
Can the preferred page be crawled, indexed, linked and used?

5 **Rights & privacy**
Are data, quotations, media and tool use authorised?

6 **Decision**
Publish, Repair or Hold—with an owner and reason.

01

TOPIC 01

Getting Started with Generative AI for SEO

SEO and generative AI · ChatGPT, Claude and Gemini · keyword research and topic clusters · search intent · content planning · effective prompting

COURSE COVERAGE First half · 3 labs

Key Concepts — Getting Started with Generative AI for SEO

1 **SEO fundamentals**
Help search engines understand a page and help people decide whether it answers their need.

2 **AI assistants**
Use ChatGPT, Claude or Gemini to transform supplied evidence, not to invent market facts.

3 **Keyword evidence**
Separate observed query data from AI suggestions and label assumptions explicitly.

4 **Topic clusters**
Group related queries around one audience problem while preventing overlapping pages.

5 **Search intent**
Infer the job behind a query, then confirm it against current result-page evidence.

6 **Structured prompting**
Use Goal, Context, Constraints, Sources, Output and Review to make quality observable.

TOPIC 01 · CONCEPT

Introduction to SEO and Generative AI

WHAT

- Search engine optimisation helps search systems understand web content and helps people decide whether a result is useful. Search discovery is not a single ranking trick: a page must be discoverable, crawlable, indexable, relevant to a need and useful after the click. Generative AI can accelerate research organisation, comparison, drafting and critique, but it does not create search demand or guarantee visibility.

WHY IT MATTERS

- A fluent AI draft can hide weak evidence. Keeping the search process and the AI workflow separate prevents common errors such as treating invented search volume as data, assuming a keyword alone determines ranking or publishing generic pages at scale. The practitioner remains responsible for the purpose, evidence, editorial quality and technical state of the page.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

TOPIC 01 · PROCESS

How It Works — Introduction to SEO and Generative AI

1. Define the audience, business purpose and page job before opening an AI assistant.

2. Collect approved business facts, query evidence and current result-page observations.

3. Use AI to organise or transform the evidence, then verify the page and monitor real outcomes.

In Practice — Introduction to SEO and Generative AI

WORKED EXAMPLE

- MerlionTrail sells repairable travel organisers. The team wants a useful guide for carry-on packing.
- Keyword evidence and current result pages suggest several related questions, but no ranking promise.
- AI organises the evidence into a plan; the human owner supplies product facts and approves the result.

DECISION GUIDE

- **Use when**

– The task has a defined user need and sources that can be supplied to the assistant.

– A human can review both the output and the live page before publication.

- **Avoid when**

– The goal is to mass-produce pages primarily to manipulate search visibility.

– The only evidence is an AI answer with no traceable source or current query data.

Quality Lens — Introduction to SEO and Generative AI

Purpose

- State the audience need and the useful outcome before choosing keywords.

Evidence

- Label business facts, observed data, hypotheses and unknowns separately.

Ownership

- Assign a human editor to approve claims, page quality and technical changes.

Setting Up ChatGPT, Claude and Gemini for SEO

WHAT

- ChatGPT, Claude and Gemini are general-purpose AI assistants with changing models and interfaces. For this course, the tool is a workspace for grounded transformation: learners provide a source pack, ask for a defined output and review it against a checklist. The course does not depend on a paid tier, API key, browser extension or automatic publishing connection.

WHY IT MATTERS

- Good setup is mostly governance. Confidential strategy, personal information, credentials and unreleased client material should not be pasted into an unapproved service. A reusable project brief, source boundary and output convention make results more consistent across tools without assuming that any vendor has unique access to Google ranking data.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

TOPIC 01 · PROCESS

How It Works — Setting Up ChatGPT, Claude and Gemini for SEO

1. Choose one approved assistant and create a fresh conversation for the C11 synthetic scenario.

2. Paste only the supplied source pack and state that missing information must be marked Unknown.

3. Save the prompt, output and human edits in local files so the reasoning trail is reviewable.

In Practice — Setting Up ChatGPT, Claude and Gemini for SEO

WORKED EXAMPLE

- A learner opens one assistant and adds the MerlionTrail source pack as delimited context.
- The assistant must cite the supplied section behind each proposed claim and label any assumption.
- The learner compares one small task in a second tool only if time allows; no account is mandatory.

DECISION GUIDE

- **Use when**

– The organisation permits the selected tool and the source material is appropriate to share.

– The output can be stored and reviewed before it affects a live page.

- **Avoid when**

– A prompt would contain passwords, unpublished client data or sensitive personal information.

– A browser add-on or automation would publish or change a website without approval.

Quality Lens — Setting Up ChatGPT, Claude and Gemini for SEO

Minimum data

- Share only the context needed for the specific SEO task.

Clear boundary

- Tell the assistant which sources it may use and how to show unknowns.

Review trail

- Keep the input, first output, edits and final decision together.

AI-Powered Keyword Research and Topic Clusters

WHAT

- Keyword research discovers the language people use around a need. A useful keyword record includes the query, source, date, location, evidence notes and an intended page job. A topic cluster groups closely related queries under a coherent primary page and supporting pages; it is an information-architecture decision, not a licence to create one near-duplicate page per phrase.

WHY IT MATTERS

- AI is good at expanding seeds, normalising wording and proposing groups, but it can fabricate volume, competition and trends. Evidence from Keyword Planner, Search Console, customer language or a current result-page review must remain distinguishable from an AI suggestion. Human review removes irrelevant phrases and resolves clusters that would compete with each other.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

TOPIC 01 · PROCESS

How It Works — AI-Powered Keyword Research and Topic Clusters

1

Start with approved products, customer questions and a small set of evidence-backed seed queries.

2

Ask AI for labelled expansions and preliminary semantic groups without inventing numeric metrics.

3

Validate relevance, combine overlapping clusters and assign one clear page job to each retained group.

In Practice — AI-Powered Keyword Research and Topic Clusters

WORKED EXAMPLE

- Seeds include 'packing cubes singapore', 'carry on packing list' and 'repair travel organiser'.
- AI proposes variants, but synthetic training metrics remain in the evidence CSV—not in the model output.
- The editor keeps three clusters: choose organisers, pack a carry-on and repair/reuse travel gear.

DECISION GUIDE

- **Use when**

– There is a real offer or audience problem and at least one traceable source of query evidence.

– Clusters can be mapped to distinct page purposes rather than wording variations alone.

- **Avoid when**

– Numeric demand or competition comes only from an AI response.

– The plan creates many thin pages whose purpose and content substantially overlap.

Quality Lens — AI-Powered Keyword Research and Topic Clusters

Source every row

- Record where the query came from and when the evidence was observed.

Separate facts

- Keep measured values, AI suggestions and human decisions in different columns.

Prevent overlap

- Give each retained cluster one primary need and one primary destination page.

Understanding Search Intent and Content Planning with AI

WHAT

- Search intent is the job a person is trying to complete. A practical planning heuristic labels a query as learn, compare, act or navigate, then records the required evidence and suitable format. Intent is inferred—not read directly from a keyword—and mixed intents are common. Current result-page features and leading page types provide observations that a human must interpret.

WHY IT MATTERS

- A page can mention a relevant phrase yet fail the user if its format and depth do not match the job. A content plan joins cluster, audience, intent, page type, unique value, evidence, internal links and success signal. AI can make this matrix faster to assemble, but it cannot safely decide business priority or claim that a result-page pattern will remain unchanged.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

TOPIC 01 · PROCESS

How It Works — Understanding Search Intent and Content Planning with AI

1. Review each cluster and state the user job in one sentence using an action verb.

2. Record current result-page observations and distinguish dominant, secondary and uncertain intent.

3. Choose a page type, unique contribution, evidence plan, internal links and measurable next action.

In Practice — Understanding Search Intent and Content Planning with AI

WORKED EXAMPLE

- 'Carry on packing list' is mainly a learn job; a checklist guide fits better than a category page.
- 'Packing cubes singapore' mixes compare and act; a category page needs selection guidance and real facts.
- Both pages can link naturally without repeating the same primary purpose.

DECISION GUIDE

- **Use when**

– The team can review current search results and articulate what a satisfied visitor would accomplish.

– The plan includes unique experience or approved evidence beyond a generic AI summary.

- **Avoid when**

– Intent is assigned from an AI label without checking the query context or result page.

– A sales page is forced onto a query whose dominant job is to learn or solve a problem.

Quality Lens — Understanding Search Intent and Content Planning with AI

User job

- Write the desired visitor outcome before selecting the page format.

SERP evidence

- Date observations and treat them as a snapshot, not a permanent rule.

Unique value

- Name the experience, example or data the page contributes.

TOPIC 01 · CONCEPT

Effective Prompting for SEO Tasks

WHAT

- A prompt is a working brief. C11 uses G-C-C-S-O-R: Goal, Context, Constraints, Sources, Output and Review. The structure tells the assistant what successful work looks like, confines it to approved evidence and requests a self-check that a human can inspect. Prompting is iterative: the user reviews a first output, identifies a specific defect and changes one instruction or source at a time.

WHY IT MATTERS

- Vague prompts produce generic copy and hide assumptions. A structured prompt reduces ambiguity, makes outputs comparable and allows a claim-to-source check. It does not remove hallucination risk, replace keyword evidence or transfer editorial accountability to the model.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

TOPIC 01 · PROCESS

How It Works — Effective Prompting for SEO Tasks

1. State one goal and the exact audience, page job and business context.

2. Add constraints, delimit approved sources and specify the output schema or example.

3. Require a review table for unsupported claims, missing evidence and guideline risks before accepting.

In Practice — Effective Prompting for SEO Tasks

WORKED EXAMPLE

- Goal: create a cluster table from the supplied CSV; Context: MerlionTrail Singapore travel gear.
- Constraints: do not invent volume; Sources: only delimited rows; Output: fixed six-column table.
- Review: list excluded terms, overlapping clusters and every statement not supported by a source row.

DECISION GUIDE

- **Use when**

– The task can be evaluated against observable criteria such as fields, sources and page purpose.

– The user is prepared to inspect and refine the output rather than accept the first response.

- **Avoid when**

– The prompt asks the model to guarantee ranking, predict proprietary metrics or conceal AI use.

– The source boundary is empty while the task requires factual claims.

Quality Lens — Effective Prompting for SEO Tasks

Specific

- Define audience, task, scope, format and the decision the output supports.

Grounded

- Delimit sources and instruct the model to show Unknown instead of inventing.

Testable

- Request a self-check, then perform an independent human review.

WHAT YOU'LL DO

Hands-On Labs — Getting Started with Generative AI for SEO

Lab 1

- Build the AI-Ready SEO Brief and Prompt Guardrails

Lab 2

- Build the Keyword Universe and Topic Clusters

Lab 3

- Map Search Intent and Create the Content Plan

TOPIC 01 · HANDS-ON

Build the AI-Ready SEO Brief and Prompt Guardrails

LAB 1

You will set up one AI assistant for the synthetic MerlionTrail scenario, distinguish approved facts from unknowns and create a reusable SEO prompt. The prompt will force the assistant to use supplied sources, label missing information and return a claim-to-source review.

You'll build

work-c11/01-ai-ready-seo-brief.md containing the audience, offer, page goal, data boundary, G-C-C-S-O-R prompt, first output and human review decision

1. Prepare the workspace
2. Extract approved facts
3. Build G-C-C-S-O-R
4. Run the evidence check

Tools: Text editor · ChatGPT, Claude or Gemini · merliontrail-brand-source-pack.md

LAB 1 · VERIFY

Build the AI-Ready SEO Brief and Prompt Guardrails

Test it

The file must contain the stated audience and page goal, exactly six approved facts, all listed unknowns, the full G-C-C-S-O-R prompt, a five-row AI output, and one evidence-based Keep, Repair and Hold decision. Search the file for '%' and 'guarantee'; neither may appear unless quoted in a rejection note.

Build the Keyword Universe and Topic Clusters

LAB 2

You will work from a synthetic keyword evidence export, preserving source and metric labels while an AI assistant proposes additional wording and preliminary groups. You will reject invented measurements, remove irrelevant terms and make the final clustering decisions yourself.

You'll build

work-c11/02-keyword-cluster-workbook.csv and work-c11/02-cluster-decisions.md with source-labelled keywords, AI suggestions, three final clusters, exclusions and overlap decisions

1. Inspect evidence
2. Label suggestions
3. Cluster by user need
4. Resolve overlap

Tools: Spreadsheet · text editor · AI assistant · synthetic-keyword-evidence.csv

LAB 2 · VERIFY

Build the Keyword Universe and Topic Clusters

Test it

The CSV must preserve every supplied source and synthetic metric, contain no numeric value on an AI_SUGGESTION row, use exactly three final cluster names and give every row a human decision and reason. The decisions file must contain at least two observed queries per cluster and at least three exclusions.

TOPIC 01 · HANDS-ON

Map Search Intent and Create the Content Plan

LAB 3

You will convert the three reviewed clusters into a content plan. The plan distinguishes dominant and secondary intent, records dated result-page observations from a supplied snapshot and assigns each page a unique job, evidence plan, internal links and success signal.

You'll build

work-c11/03-intent-content-plan.csv with one approved row per cluster plus a short cannibalisation boundary and human decision for each planned page

1 **State the user job**

2 **Record SERP clues**

3 **Choose page type**

4 **Plan links and evidence**

Tools: Spreadsheet · AI assistant · cluster decisions · synthetic-serp-observations.csv

LAB 3 · VERIFY

Map Search Intent and Create the Content Plan

Test it

The CSV must contain exactly three planned pages, the three specified primary queries, all three aggregated SERP clues per cluster, the supplied observation date, Intent_Review and Human_Resolution, a verb-led page job, planned URL, Product Editor owner, traceable unique contribution, descriptive anchor text in both directions, success signal, overlap boundary and human decision. No cell may promise a position, traffic amount or guaranteed result.

TOPIC RECAP

Recap - Getting Started with Generative AI for SEO

1. LO1 - Set Up and Prompt: Explain the SEO and generative-AI relationship, keep evidence separate from suggestions, and use G-C-C-S-O-R with a human review gate.

2. LO2 - Research and Plan: Build labelled keyword evidence, resolve three topic clusters, and map each cluster to a distinct intent-led page with dated clues and descriptive internal links.

02

TOPIC 02

Creating and Optimising SEO Content with Generative AI

Titles and meta descriptions · briefs, outlines and long-form content · on-page and technical SEO · page audits · fact-checking · Google guidance

COURSE COVERAGE Second half · 3 labs

Key Concepts — Creating and Optimising SEO Content with Generative AI

1. **Search appearance**
   Write descriptive title and meta preferences while recognising that Google may generate alternatives.

2. **Content brief**
   Join audience, intent, evidence, unique value, structure, links and acceptance criteria.

3. **People-first drafting**
   Use AI for structure and options; add original experience, accurate facts and editorial judgment.

4. **On-page signals**
   Align visible title, headings, body, images and internal links around one clear page job.

5. **Technical triage**
   Check indexability, canonical choice, crawlable links, mobile usability and page experience.

6. **Quality control**
   Verify every material claim, disclose automation where useful and avoid scaled low-value production.

Generating Titles, Meta Descriptions and On-Page Content

WHAT

- A page title is a preference used among several signals when Google creates a title link. A meta description is a page-specific summary that Google may use when it better describes the page than on-page text. Neither is a fixed-length ranking formula. On-page content should clearly answer the visitor's job with a distinct main heading, useful sections and natural language.

WHY IT MATTERS

- AI can create alternatives quickly, but generic formulas often cause boilerplate titles, repeated descriptions and keyword stuffing. The editor should choose wording that accurately represents the page, differentiates it from other site pages and sets the right expectation before the click.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Generating Titles, Meta Descriptions and On-Page Content

1. Write the page job and primary topic, then draft several descriptive and concise title options.

2. Create a unique meta summary using only facts actually present on that page.

3. Check title, visible H1, opening, sections and call to action for one coherent promise.

In Practice — Generating Titles, Meta Descriptions and On-Page Content

WORKED EXAMPLE

- Weak title: 'Packing Cubes, Packing Cube, Best Packing Cubes | MerlionTrail'.
- Improved preference: 'How to Choose Packing Cubes for Carry-On Travel | MerlionTrail'.
- The description summarises the selection guide and repairable construction without a ranking claim.

DECISION GUIDE

- **Use when**

– The page has a distinct purpose and approved facts that can support a useful summary.

– Alternative wording will be reviewed against the actual visible content.

- **Avoid when**

– Titles repeat phrases, add unsupported superlatives or differ materially from the page.

– One generic description is copied across every page.

Quality Lens — Generating Titles, Meta Descriptions and On-Page Content

Accurate

- The title, H1, snippet preference and page content describe the same primary job.

Distinct

- Wording differentiates this URL from other site pages.

Natural

- Use human-readable language and avoid repeated keyword variants.

TOPIC 02 · CONCEPT

Writing Briefs, Outlines and Long-Form Content with AI

WHAT

- A content brief translates research into an editorial contract: audience, user job, primary topic, unique value, approved sources, coverage, exclusions, internal links, conversion path and review criteria. An outline sequences the answer. A draft turns that plan into readable content while retaining source boundaries and the organisation's genuine experience.

WHY IT MATTERS

- Starting with 'write an SEO article' encourages generic prose and invented facts. Brief-first generation lets the human settle strategy before wording. It also creates checkpoints: an outline can be rejected cheaply, claims can be traced and the draft can be evaluated for completeness without using word count as a proxy for usefulness.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

TOPIC 02 · PROCESS

How It Works — Writing Briefs, Outlines and Long-Form Content with AI

1

Complete the brief from the approved cluster, intent plan, source pack and unique contribution.

2

Generate an outline with a one-sentence purpose for every section and remove duplicated coverage.

3

Draft section by section, preserving citations and adding human examples, edits and a final read-through.

In Practice — Writing Briefs, Outlines and Long-Form Content with AI

WORKED EXAMPLE

- The carry-on guide promises a printable seven-step checklist and a repair/reuse decision table.
- The outline moves from bag constraints to categories, packing order, organiser choice and a final check.
- The draft uses only supplied product facts; general travel rules stay out unless independently sourced.

DECISION GUIDE

- **Use when**

– The team can state what is uniquely useful and which sources support the page.

– A human editor will review outline, claims, voice and final flow.

- **Avoid when**

– The only brief is a target word count and a keyword-density request.

– The model is asked to imitate a competitor or rewrite copyrighted text closely.

Quality Lens — Writing Briefs, Outlines and Long-Form Content with AI

Brief before prose

- Approve the user job, evidence and unique value before drafting.

One section, one job

- Give each heading a clear question or decision to resolve.

Human contribution

- Add first-hand detail, examples or judgment that the model cannot supply.

On-Page and Technical SEO Optimisation with AI

WHAT

- On-page SEO covers the visible and embedded signals that explain one page: title, main heading, sections, media text and internal links. Technical SEO ensures search systems can access and process the intended version: crawlable links, an indexable response, a sensible canonical, mobile usability and acceptable page experience. AI can explain code and propose a checklist, but the site owner must inspect the real page and approve changes.

WHY IT MATTERS

- A strong article can remain hard to discover if it is blocked, orphaned or duplicated; technical perfection cannot rescue unhelpful content. A layered check—purpose, on-page, crawl/index and experience—keeps the audit proportional and prevents automated tools from producing an unprioritised list of warnings.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — On-Page and Technical SEO Optimisation with AI

1

Confirm the preferred URL loads, returns indexable content and has a consistent title and main heading.

2

Review descriptive crawlable internal links, image text, canonical choice and accidental robots directives.

3

Record page-experience observations, assign an owner and verify changes with the appropriate live tool.

In Practice — On-Page and Technical SEO Optimisation with AI

WORKED EXAMPLE

- The sample page has a vague title, two H1 elements, a noindex directive and a JavaScript-only link.
- Content edits alone will not solve the noindex issue; the release owner must remove it deliberately.
- The corrected internal link uses a real anchor element and descriptive text to a related packing guide.

DECISION GUIDE

- **Use when**

– The practitioner has authority to inspect the page and can involve a developer for release changes.

– Each finding is tied to impact, evidence, owner and a verification method.

- **Avoid when**

– AI-generated code is pasted into production without review, backup and testing.

– Tool scores are treated as ranking guarantees or every warning receives equal priority.

Quality Lens — On-Page and Technical SEO Optimisation with AI

Layered audit

- Check page purpose, on-page signals, crawl/index state and experience separately.

Evidence

- Record the element, URL or tool observation behind each finding.

Release control

- Assign technical changes to an authorised owner and verify after deployment.

Auditing and Improving Existing Pages

WHAT

- A page audit compares the page's intended job, current evidence and observed performance with a clear standard. Useful findings describe the problem, evidence, likely user or search impact, recommended change, owner and verification step. Improvement is a controlled cycle: establish a baseline, diagnose, prioritise, change, verify and monitor.

WHY IT MATTERS

- AI can summarise a page and detect inconsistencies, but it lacks proprietary Search Console data unless supplied and may overstate causes. Search performance changes for many reasons. A disciplined audit separates direct observations from hypotheses and avoids rewriting an entire page before the team knows which problem it is solving.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

TOPIC 02 · PROCESS

How It Works — Auditing and Improving Existing Pages

1

Record the intended query cluster, page job and baseline clicks, impressions, CTR and position context.

2

Inspect content, search appearance, links and technical state; label facts, hypotheses and unknowns.

3

Prioritise by user impact, confidence and effort, implement one coherent change set and monitor.

In Practice — Auditing and Improving Existing Pages

WORKED EXAMPLE

- The sample page has impressions but weak clicks for carry-on queries and a title that says only 'Products'.
- The title mismatch is an observation; its contribution to CTR is a hypothesis to test after correction.
- The plan fixes indexability first, then aligns the title/H1 and improves the guide with approved evidence.

DECISION GUIDE

- **Use when**

– There is a defined page purpose and baseline evidence can be preserved.

– The team can observe the page after changes and avoid changing unrelated variables at the same time.

- **Avoid when**

– A single metric movement is attributed to one cause without sufficient evidence.

– An AI audit is used as a direct production-change list without human triage.

Quality Lens — Auditing and Improving Existing Pages

Observation

- Capture what the page, source code or report actually shows.

Hypothesis

- State the suspected effect and confidence without presenting it as fact.

Verification

- Define what will be checked, by whom and over what observation window.

Fact-Checking, Quality Control and Google Guidance

WHAT

- Quality control verifies claims, sources, originality, usefulness, language, rights, privacy and technical readiness. Google's guidance focuses on helpful, reliable, people-first content and asks creators to consider who made it, how it was made and why it exists. Appropriate AI assistance is not automatically prohibited; using automation primarily to manipulate rankings or create scaled low-value content can violate spam policies.

WHY IT MATTERS

- Generative systems can produce confident errors, outdated statements and unsupported comparisons. An editorial gate makes every material claim traceable and gives the editor a deliberate Publish, Repair or Hold decision. Disclosing automation may give readers useful context when they would reasonably expect to know how the material was created.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

TOPIC 02 · PROCESS

How It Works — Fact-Checking, Quality Control and Google Guidance

1

Build a claim ledger and verify each material statement against an authoritative or approved source.

2

Review originality, audience value, Who-How-Why context, rights, privacy and search-policy risks.

3

Run the on-page and technical checks, record the human owner and choose Publish, Repair or Hold.

In Practice — Fact-Checking, Quality Control and Google Guidance

WORKED EXAMPLE

- The AI draft says MerlionTrail products 'cut packing time by 40%' although no source supports it.
- The editor removes the claim, retains the approved repairable-material facts and records the source.
- The page is held until its noindex directive is removed and the final HTML is checked.

DECISION GUIDE

- **Use when**

– Every important claim can be traced, corrected or explicitly marked as unknown.

– A named human owner can explain the page's purpose and approve publication.

- **Avoid when**

– Unsupported claims are kept because they sound plausible or include a desirable keyword.

– Large batches of near-duplicate pages are generated without original value or editorial review.

Quality Lens — Fact-Checking, Quality Control and Google Guidance

Truth

- Trace claims to sources and remove or qualify anything unsupported.

Value

- Confirm the page adds useful, original contribution for its intended audience.

Readiness

- Approve content, rights, privacy, search-policy and technical checks together.

WHAT YOU'LL DO

Hands-On Labs — Creating and Optimising SEO Content with Generative AI

Lab 4

- Create the SEO Brief, Outline, Metadata and First Draft

Lab 5

- Audit and Optimise the Existing Page

Lab 6

- Run the Quality Gate and Build the SEO Improvement Plan

Create the SEO Brief, Outline, Metadata and First Draft

LAB 4

You will turn the approved Pack a Carry-On plan into a content brief, outline, title preference, meta description, opening and checklist. The assistant may use only supplied sources, and every factual claim must carry a source label for later verification.

You'll build

work-c11/04-content-package.md containing the approved brief, outline, three title options, two meta descriptions, a selected opening and checklist, claim labels and a human edit log

1 Complete the brief

2 Approve the outline

3 Draft metadata

4 Generate grounded copy

Tools: Text editor - AI assistant - content-brief-template.md - prior lab checkpoints

Create the SEO Brief, Outline, Metadata and First Draft

Test it

The package must have every brief field, a five-to-seven-section outline with no UNKNOWN source kept, three title options, two descriptions, one selected title and description, a selected 100-150 word opening, a 350-500 word checklist, a claim ledger and at least three human edits. Prohibited superlatives and ranking promises must be absent.

TOPIC 02 · HANDS-ON

Audit and Optimise the Existing Page

LAB 5

You will inspect a deliberately flawed local HTML page. The audit separates page purpose, on-page signals, crawl/index state and experience. You will record evidence for every finding, correct the authorised local copy, preserve a before/after diff and keep production deployment as a separate human-controlled action.

You'll build

work-c11/05-page-audit.csv, work-c11/05-optimised-page.html and work-c11/05-page-changes.diff with evidence, priorities, authorised corrections and an explicit verification record

1 **Baseline the page**

2 **Audit four layers**

3 **Prioritise findings**

4 **Correct and verify**

Tools: Browser - text editor - spreadsheet - AI assistant - existing-carry-on-page.html

Audit and Optimise the Existing Page

Test it

The audit must contain at least ten evidence-backed findings across all four layers, one priority, owner and verification per row. The corrected HTML must pass every checklist item and open locally. The diff must show only changes represented in the audit, and head values must be verified through source or developer tools. No row may promise ranking or traffic.

TOPIC 02 · HANDS-ON

Run the Quality Gate and Build the SEO Improvement Plan

LAB 6

You will complete the workflow with a claim ledger, people-first content review, synthetic performance baseline and release decision. The final plan separates observations from hypotheses and states which owner will verify content and technical changes before any live publication.

You'll build

work-c11/06-quality-gate.md, work-c11/06-performance-baseline.csv and work-c11/06-action-plan.csv with claim evidence, Who-How-Why review, calculated CTR, three actions, guardrails and a final decision

1. **Build the claim ledger**
2. **Apply Who-How-Why**
3. **Review performance evidence**
4. **Decide and monitor**

Tools: Text editor - spreadsheet - AI assistant - corrected HTML - synthetic-search-console.csv

Run the Quality Gate and Build the SEO Improvement Plan

Test it

The quality gate must contain a complete claim ledger, Who-How-Why review, rights/privacy/scaled-content checks, dated official guidance citations and the required Repair decision. The performance baseline must preserve six synthetic rows and show CTR values 1.50%, 1.00%, 1.33%, 2.30%, 1.43% and N/A. The separate action plan must contain exactly three rows with priority, owner, window, success signal and guardrail. Observations and hypotheses must be visibly separate.

TOPIC RECAP

Recap - Creating and Optimising SEO Content with Generative AI

1. LO3 - Create and Optimise: Turn the approved intent plan into a source-grounded brief, outline, metadata, opening and checklist with a claim ledger and human edits.

2. LO4 - Audit and Verify: Audit purpose, on-page, crawl/index and experience layers; correct the local page; and make a source-backed Repair decision with owners, metrics and guardrails.

WRAP-UP

Course Summary & Next Steps

LEARNING OUTCOMES

What You Achieved

1

Set Up & Prompt

Explain how SEO and generative AI work together, set safe working boundaries and use a structured prompt-and-review method.

2

Research & Plan

Build an evidence-led keyword universe, topic clusters and a search-intent content plan with AI assistance.

3

Create & Optimise

Create an SEO content brief, outline, title, meta description and on-page draft that remain grounded in approved sources.

4

Audit & Verify

Audit and improve an existing page for content, on-page and technical SEO, then verify quality against Google guidance.

NEXT STEPS

Continue the Workflow

1. Reuse the prompt, research and quality-control templates on an authorised page.

2. Refresh query and result-page evidence before making a new content decision.

3. Monitor outcomes in Search Console and distinguish observations from hypotheses.

4. Keep a human owner accountable for every page change and publication decision.

WRAP-UP

Keep Practising After the Course

1. Re-run the labs on your own machine — repetition is what makes the skills stick.
2. Apply the techniques to a real process or project in your own organisation.
3. Keep your downloaded C11 course files and use the lab checkpoints to restart safely.
4. Explore the follow-on courses at www.tertiarycourses.com.sg.

GENERATIVE AI FOR SEO · C11

Thank You!

You can now research, create, optimise and verify SEO content with generative AI while keeping evidence, useful purpose and human judgment in control.